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SET C

Transformer Mechanics Worksheet

Question 1: Tokenization (5 marks). The word "MISUNDERSTANDING" is uncommon enough that a sub-word tokenizer would likely split it into smaller pieces rather than keep it whole.

(a) Split "MISUNDERSTANDING" into 3 reasonable sub-word pieces (there's no single "correct" answer—justify your split).

Answer :-

MIS + UNDER + STANDING

Justification:

- MIS – A common prefix meaning "wrongly" or "incorrectly" (e.g., misbehave, misplace).
- UNDER – A frequently occurring English word and prefix used in many contexts.
- STANDING – A common word derived from "stand" with the suffix "-ing," indicating a state or action.

A sub-word tokenizer (such as Byte Pair Encoding or WordPiece) typically breaks uncommon words into smaller, meaningful units that appear frequently in the training data. Therefore, splitting "MISUNDERSTANDING" into these three parts is a reasonable approach.

(b) In 1–2 sentences, explain WHY sub-word tokenization (rather than whole-word tokenization) helps a model handle a word like this one, even if the model has never seen the whole word before in training.

Answer :-

Sub-word tokenization allows language models to represent rare or unseen words by breaking them into smaller, familiar components. Instead of treating an unknown word as a single token, the model can understand it using the meanings and patterns of its sub-words.

For example, even if the model has never encountered the complete word "MISUNDERSTANDING", it may have seen "MIS", "UNDER", and "STANDING" many

times during training. By combining the information from these known pieces, the model can infer that the word refers to an incorrect understanding or confusion.

This approach:

1. Reduces the number of unknown words.
2. Improves vocabulary efficiency.
3. Helps the model generalize to new and complex words.
4. Enhances performance on languages with many word variations.

Hence, sub-word tokenization makes transformer models more flexible and effective when handling uncommon or previously unseen words.

Question 2: Embeddings (5 marks). Consider the word pairs "happy" and "joyful" and "happy" and "chair".

(a) Which pair would you expect to have HIGHER cosine similarity in embedding space, and why?

Answer :-

The pair "happy" and "joyful" would have a higher cosine similarity than "happy" and "chair".

Why:

- Semantic Relation: "Happy" and "joyful" are near-synonyms describing similar positive emotional states, so they frequently appear in similar contexts within text corpora.
- Vector Direction: Embedding algorithms map semantically related words closer together in vector space. Because their vectors point in nearly the same direction, the cosine of the angle between them is close to 1.
- By contrast, "happy" (an emotion) and "chair" (a physical object) share virtually no contextual overlap, resulting in nearly orthogonal vectors with a cosine similarity close to 0.

(b) In your own words, what does an embedding actually represent, and how is it different from a one-hot vector?

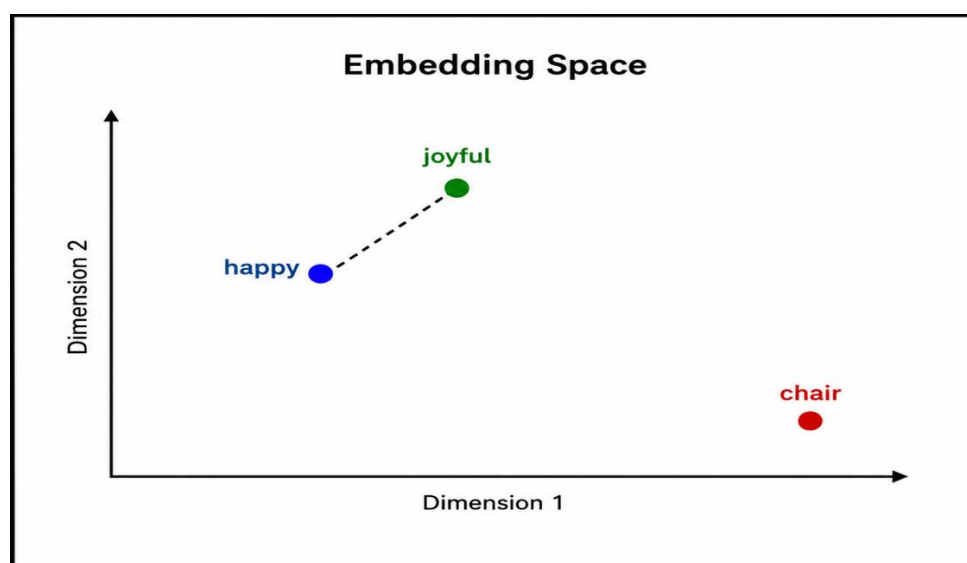
Answer :-

What an Embedding Represents:

An embedding is a dense, continuous numerical vector that represents the semantic and syntactic meaning of a word in a lower-dimensional vector space. Rather than treating words as arbitrary symbols, an embedding captures real-world relationships—allowing similar concepts to cluster close together in space.

Key Differences :-

Feature	One-Hot Vector	Embedding Vector
Structure	Sparse: Contains all 0s with a single 1 at the word's index position.	Dense: Contains learned continuous floating-point values across all dimensions.
Dimensionality	Extremely high (equal to total vocabulary size V , e.g., 50,000+).	Compact and fixed (typically $d \in [256, 1024]$ dimensions).
Semantics	None: Every word is equidistant from every other word; dot product between any two distinct words is 0.	Rich: Similar words have high dot products/cosine similarity, capturing analogies and concepts.



Question 3: Self-Attention Worked Example (5 marks). Read the sentence: "Sam tried to call George on the phone, but he wasn't successful."

(a) The word "he" should attend most strongly to which other word in the sentence? Circle one:

- "Sam"
- "George"

Answer :-

Sam

Explanation:

In the sentence, "Sam tried to call George on the phone, but he wasn't successful," the person performing the action is Sam. The phrase "wasn't successful" refers to the attempt to make the phone call. Since Sam was the one trying to call George, it is logical that "he" refers to Sam. Therefore, the self-attention mechanism should assign a stronger attention weight between "he" and "Sam" than between "he" and "George."

(b) Explain in 2–3 sentences WHY self-attention (not a simple word-order rule) is needed to correctly resolve what "he" refers to here.

Answer :-

A simple word-order rule—such as selecting the nearest noun or the most recent subject—fails here because it would incorrectly map "he" to "George" simply due to spatial proximity.

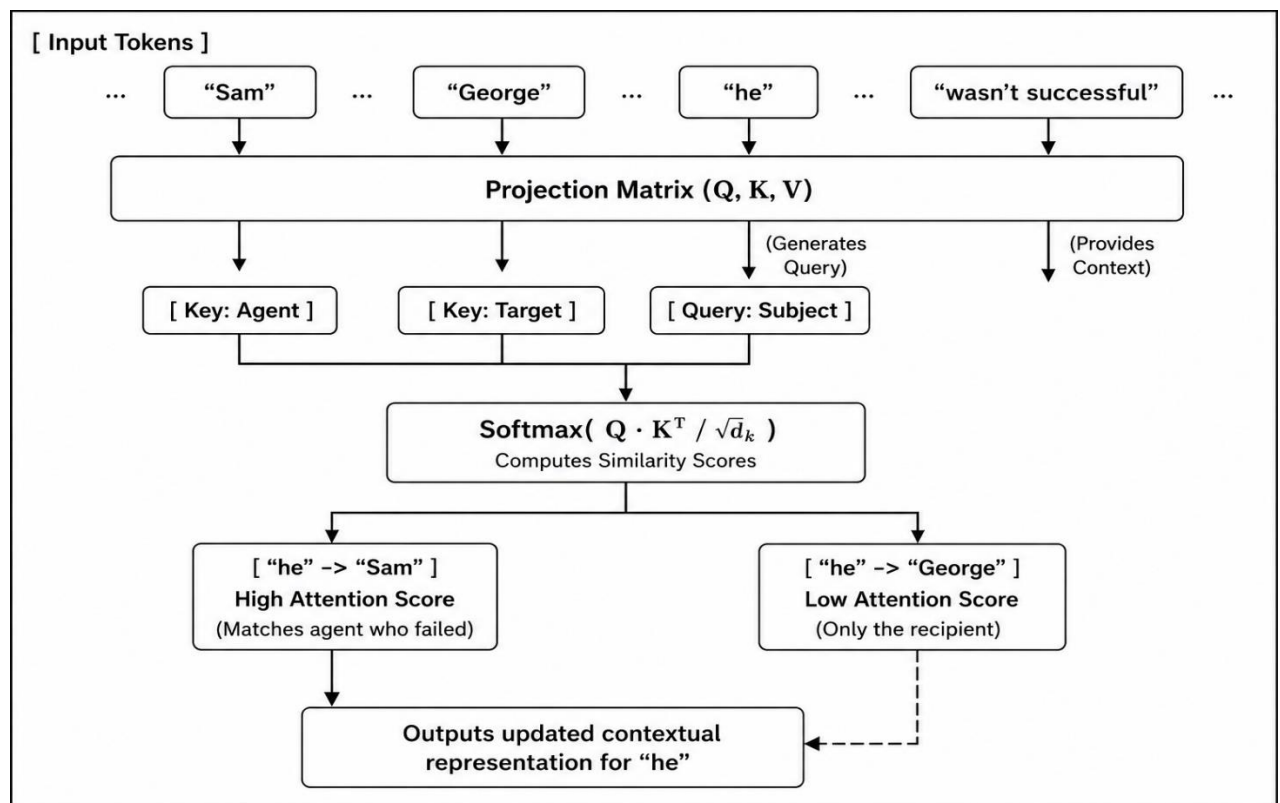
Self-attention avoids this trap by evaluating contextual semantics rather than physical distance:

- **Contextual Role Matching:** The clause "wasn't successful" modifies the outcome of an attempted action. Because "Sam" is established as the agent attempting the action ("Sam tried to call..."), the model's Query vector for "he" matches most strongly with the Key vector for "Sam".
- **Global Context Fusion:** Through self-attention matrix operations, the token "he" attends to the entire sentence simultaneously. It weighs the semantic link between *trying an action* and *failing at that action*, correctly assigning higher attention weight to "Sam" over "George" (the recipient of the call).

Visual Summary of Attention Weights :-

Token Pair	Distance in Sentence	Semantic Connection	Attention Weight
"he" → "George"	2 words (Very close)	Recipient of call attempt	Low
"he" → "Sam"	8 words (Farther)	Agent who tried and wasn't successful	High (Selected)

Transformer self attention process :-



Conclusion

1. Self-attention helps Transformer models determine the correct meaning of words by analyzing the entire sentence context rather than relying on word position alone.
2. In this example, it correctly identifies that "he" refers to "Sam", despite "George" being closer in the sentence.

Question 4: Query, Key, Value (5 marks). In your own words, briefly explain what the Query, Key, and Value vectors each represent in self-attention, and how they combine to produce an attention score.

Answer :-

What Query, Key, and Value Vectors Represent :-

- **Query (Q):** Represents the current token's search request. It asks: "What kind of information or context am I looking for in other tokens?"
- **Key (K):** Represents the indexing label or identity of each token in the sequence. It asks: "What kind of information or role do I contain that others might need?"
- **Value (V):** Represents the actual content or information carried by each token. Once attention is determined, the Value vector is what actually gets extracted and passed forward to update the representation.

How They Combine to Produce Attention Scores & Final Output

1. **Similarity Score (Dot Product):** For a target token, its Query vector (Q) is multiplied by the transpose of the Key vectors (K^T) of all tokens:

$$\text{Raw Score} = Q \cdot K^T$$

2. **Scaling:** The dot product is divided by $\sqrt{d_k}$ (where d_k is the dimension of the key vectors) to stabilize gradients during training.
3. **Softmax Normalization:** A Softmax function is applied to convert these raw similarity scores into probabilities (weights between 0 and 1 that sum to 1):

$$\text{Attention Weights} = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right)$$

4. **Weighted Value Sum:** Finally, these attention weights are multiplied by the corresponding **Value vectors (V)** to create the final contextual output vector:

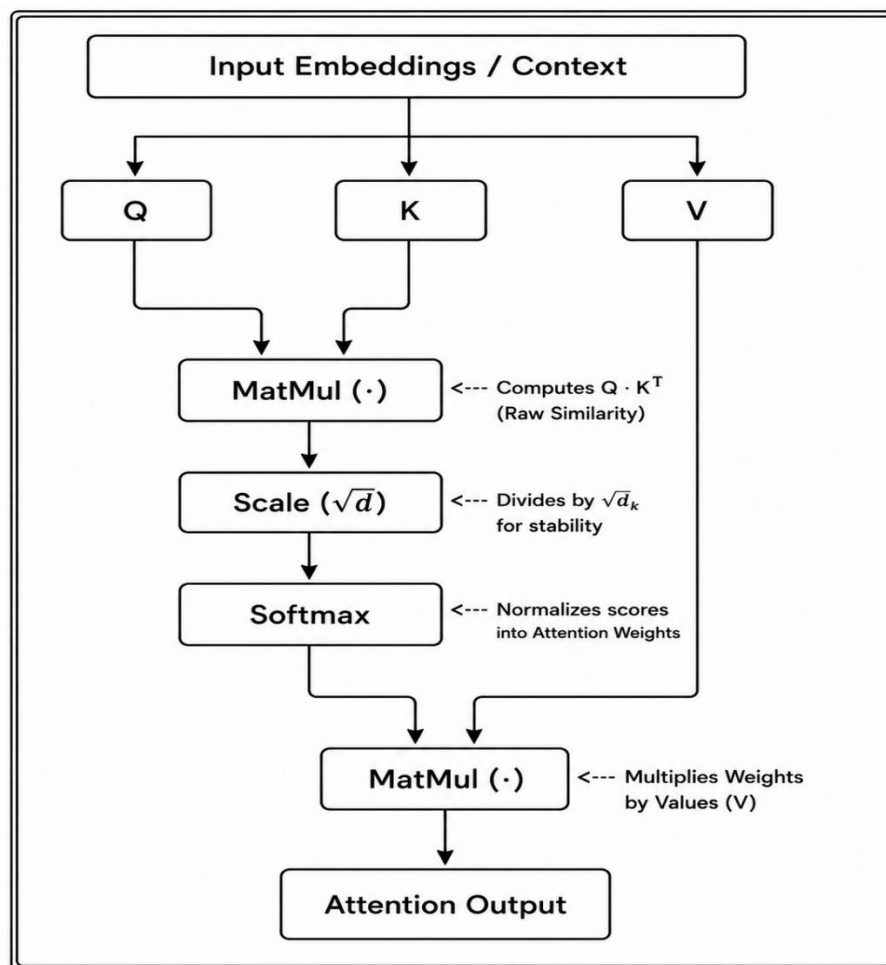
$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{Q \cdot K^T}{\sqrt{d_k}}\right) V$$

Analogy Breakdown :-

Think of it like a search engine or database lookup:

- Query: What you type in the search bar.
- Key: The tags or titles assigned to each web page/document.
- Value: The actual text content inside the web page you read once you click the best match.

Scaled Dot-Product Attention Architecture :-



Conclusion

1. The self-attention mechanism enables Transformer models to understand relationships between words by using Query, Key, and Value vectors.
2. By computing attention scores, the model identifies the most relevant contextual information in a sentence.